

Table 2: Detection results on **PASCAL VOC 2007 test set** (trained on VOC 2007 trainval). The detectors are Fast R-CNN with ZF, but using various proposal methods for training and testing.

train-time region proposals		test-time region proposals		mAP (%)
method	# boxes	method	# proposals	
SS	2000	SS	2000	58.7
EB	2000	EB	2000	58.6
RPN+ZF, shared	2000	RPN+ZF, shared	300	59.9
<i>ablation experiments follow below</i>				
RPN+ZF, unshared	2000	RPN+ZF, unshared	300	58.7
SS	2000	RPN+ZF	100	55.1
SS	2000	RPN+ZF	300	56.8
SS	2000	RPN+ZF	1000	56.3
SS	2000	RPN+ZF (no NMS)	6000	55.2
SS	2000	RPN+ZF (no cls)	100	44.6
SS	2000	RPN+ZF (no cls)	300	51.4
SS	2000	RPN+ZF (no cls)	1000	55.8
SS	2000	RPN+ZF (no reg)	300	52.1
SS	2000	RPN+ZF (no reg)	1000	51.3
SS	2000	RPN+VGG	300	59.2

Some RPN proposals highly overlap with each other. To reduce redundancy, we adopt non-maximum suppression (NMS) on the proposal regions based on their *cls* scores. We fix the IoU threshold for NMS at 0.7, which leaves us about 2000 proposal regions per image. As we will show, NMS does not harm the ultimate detection accuracy, but substantially reduces the number of proposals. After NMS, we use the top- N ranked proposal regions for detection. In the following, we train Fast R-CNN using 2000 RPN proposals, but evaluate different numbers of proposals at test-time.

4 EXPERIMENTS

4.1 Experiments on PASCAL VOC

We comprehensively evaluate our method on the PASCAL VOC 2007 detection benchmark [11]. This dataset consists of about 5k trainval images and 5k test images over 20 object categories. We also provide results on the PASCAL VOC 2012 benchmark for a few models. For the ImageNet pre-trained network, we use the “fast” version of ZF net [32] that has 5 convolutional layers and 3 fully-connected layers, and the public VGG-16 model⁷ [3] that has 13 convolutional layers and 3 fully-connected layers. We primarily evaluate detection mean Average Precision (mAP), because this is the actual metric for object detection (rather than focusing on object proposal proxy metrics).

Table 2 (top) shows Fast R-CNN results when trained and tested using various region proposal methods. These results use the ZF net. For Selective Search (SS) [4], we generate about 2000 proposals by the “fast” mode. For EdgeBoxes (EB) [6], we generate the proposals by the default EB setting tuned for 0.7

IoU. SS has an mAP of 58.7% and EB has an mAP of 58.6% under the Fast R-CNN framework. RPN with Fast R-CNN achieves competitive results, with an mAP of 59.9% while using *up to* 300 proposals⁸. Using RPN yields a much faster detection system than using either SS or EB because of shared convolutional computations; the fewer proposals also reduce the region-wise fully-connected layers’ cost (Table 5).

Ablation Experiments on RPN. To investigate the behavior of RPNs as a proposal method, we conducted several ablation studies. First, we show the effect of sharing convolutional layers between the RPN and Fast R-CNN detection network. To do this, we stop after the second step in the 4-step training process. Using separate networks reduces the result slightly to 58.7% (RPN+ZF, unshared, Table 2). We observe that this is because in the third step when the detector-tuned features are used to fine-tune the RPN, the proposal quality is improved.

Next, we disentangle the RPN’s influence on training the Fast R-CNN detection network. For this purpose, we train a Fast R-CNN model by using the 2000 SS proposals and ZF net. We fix this detector and evaluate the detection mAP by changing the proposal regions used at test-time. In these ablation experiments, the RPN does not share features with the detector.

Replacing SS with 300 RPN proposals at test-time leads to an mAP of 56.8%. The loss in mAP is because of the inconsistency between the training/testing proposals. This result serves as the baseline for the following comparisons.

Somewhat surprisingly, the RPN still leads to a competitive result (55.1%) when using the top-ranked

8. For RPN, the number of proposals (e.g., 300) is the maximum number for an image. RPN may produce fewer proposals after NMS, and thus the average number of proposals is smaller.